How Do Passengers Perceive Turbulence Risk Compared to Actual Forecast Accuracy?

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Abstract:

This study examines the perceived risk of turbulence among airline travelers and the reliability of contemporary forecasting systems. Despite technological innovations represented by Doppler radar and Eddy Dissipation Rate (EDR) modeling, a large number of travelers still regard turbulence as unpredictable or hazardous. Through the application of survey research and a regional case study of the Colorado Rockies, this study identifies the disparity between public perception and meteorological accuracy. Evidence suggests that improvements in forecasting have not been proportionate to public perception, stressing the need for improved communication between air transport professionals and travelers.

Introduction:

Atmospheric turbulence is a leading cause of in-flight discomfort and persistent source of anxiety among airline passengers. Although turbulence is a global phenomenon, it is especially prevalent in regions with mountainous terrain, such as Colorado, where wind patterns can intensify its effects. Despite the infrequency of turbulence-related injuries and accidents, public concern regarding their severity remains high (Weisbrot, 2025). Advancements in detection technologies—such as Doppler radar systems and the European Centre for Medium-Range Weather Forecasts (ECMWF) Integrated Forecasting System (IFS)—have significantly improved the precision of turbulence forecasting, as discussed by Sharman, Uhlenbrock, and Gisinger (2016; 2007; 2024). Moreover, recent studies have demonstrated that machine learning techniques and metrics, such as the Doppler spectrum width, allow for high-resolution, scale-dependent

turbulence detection (Majewski, 2024). However, there is limited public awareness of these innovations, which may contribute to the ongoing mistrust in forecast reliability.

Thus, this research explores whether modern turbulence forecasting tools, despite their proven accuracy, are effectively misunderstood and distrusted by passengers. By examining how individuals interpret turbulence risk during flight, this study aims to uncover the psychological and communicative gaps between technical forecasting systems, commercial pilot protocols, and public perceptions. By drawing on survey data from airline passengers, a regional case study centered in Colorado, and atmospheric model outputs to assess the degree of alignment between the perceived and actual turbulence risk. The findings aim to inform strategies to improve public understanding and strengthen trust in aviation meteorology and related industries.

Literature Review:

To better understand this disconnect, it is important to define turbulence and how it functions within the atmosphere. Turbulence, or "eddying," is a random, unstable air flow with abrupt, random velocity, pressure, and direction changes between air masses. It is caused by many atmospheric differentials and is described numerically by fluid mechanics (Gisinger, 2024), specifically through parameters such as the Reynolds number (Re), which is a fundamental framework of aerodynamic theory expressing the ratio of inertial to viscous forces of a fluid. A large Reynolds number indicates turbulent flow regimes, where air flows in a disordered manner owing to rivalry between various forces. However, most mathematical concepts for turbulence, if any, are too complex and give rise to whether we lean on a "philosophical approach in the Laws of Motion" (Sutton, 2019).

Turbulence is caused by physical processes, such as viscosity, conduction, and diffusion, which create molecular movement and mixing on macroscopic scales. Although turbulence can be represented using advanced mathematical models, the inherent randomness of atmospheric motion often resists precise prediction, highlighting the need for continued research that extends beyond deterministic frameworks. Persistent theoretical limitations may hinder public understanding and increase passenger anxiety.

According to Insider Tech, turbulence has not caused a plane crash in over 40 years, and improvements in aircraft engineering and aerodynamics have significantly reduced the risk of structural damage from turbulence (2022). However, turbulence is becoming increasingly frequent and intense. Since 1958, turbulence has increased by 40–90% over Europe and North America, an alarming trend that researchers increasingly link to climate change. Atmospheric scientist Dr. Paul D. Williams, a professor at the University of Reading and leading expert in aviation meteorology, has conducted extensive research on this topic. In a widely cited study, Williams demonstrated that rising atmospheric instability driven by greenhouse gas emissions strengthens jet stream wind shears, particularly over the North Atlantic. His findings suggest that light, moderate, and severe clear air turbulence are projected to increase in both frequency and intensity over major flight corridors (2017). This indicates that climate change may not only intensify turbulence but also pose growing operational and safety challenges for commercial aviation flights decades ahead.

While global trends indicate an increase in clear air turbulence due to climate change (Williams, 2017), regional topography significantly contributes to localized turbulence events. In Colorado, the complex interplay between mountainous terrain and

severe weather conditions presents a unique challenge for aviation. Dr. T. L. Keller, affiliated with NOAA's Environmental Research Laboratories in Boulder, Colorado, has extensively studied this phenomenon. In a detailed case study, Keller et al. analyzed a commercial jetliner incident near Denver International Airport, attributing the event to intense lee waves and turbulent air patterns formed when stable airflow is disrupted by mountainous terrain. Their research highlighted the dangers of vertical wind shear and strong wave activity, even under clear-sky conditions, emphasizing the need for improved detection and forecasting methods (2015).

Expanding on this research, Dr. Maria Vittoria Guarino, an atmospheric scientist and numerical modeler at the International Center for Theoretical Physics (ICTP) of the British Antarctic Survey, has delved into the dynamics of mountain-wave turbulence. Guarino et al. investigated how directional wind shear amplifies mountain-wave turbulence over the Rockies. Their findings confirmed that the interaction between topography and wind structure can intensify vertical air motion, often beyond the detection capabilities of standard forecasting models (2018). Together, these studies indicate a critical need for enhanced regional forecasting models and greater public awareness of the unique turbulence risks associated with mountainous regions, such as Colorado.

James Duncan Jr., an atmospheric scientist at Texas Tech University, conducted a study alongside colleagues to examine the limitations of Doppler radar in detecting atmospheric turbulence. In their 2019 research on spatial turbulence intensity, Duncan et al. found that standard radar systems—particularly those mounted on fixed meteorological towers—struggle to capture real-time turbulence due to limited revisit

times. These delays reduce the radar's ability to track fast-changing atmospheric conditions, making current forecasting methods less reliable for detecting short-lived or small-scale turbulence events. This technological gap may contribute to why turbulence still appears unpredictable to passengers and flight crews, despite the presence of modern radar systems. Previously, the Doppler spectrum width was considered unreliable for turbulence detection owing to spectral broadening (caused by aircraft motion, wind shear, and hydrometeor motion).

Other studies have attempted to validate the accuracy of the spectral width by comparing it with established turbulence metrics, such as the eddy dissipation rate (EDR) and gust probe data from research aircraft. More recently, Doppler radars have expanded their forecasting, as they found that turbulence varies based on altitude, atmospheric stability, and wind conditions (Majewski et. al, 2023). Traditional turbulence forecasting relies on kilometer-scale models; however, this new approach can detect turbulence at finer scales (~10-30 meters). Higher-resolution turbulence mapping can help pilots avoid dangerous turbulence zones more efficiently.

A new technique (STI) was developed to measure turbulence spatially, rather than relying solely on temporal wind fluctuations. The STI helps create turbulence maps showing intensity in different regions, which could improve turbulence forecasting for pilots and aviation meteorologists. STI validation has some limitations, such as measurement timing differences (between radar and ground stations) and volume-averaging errors. As this forecasting model evolved, the STI successfully captured rapid turbulence fluctuations, which helped forecast turbulence intensity more effectively than previous models in the aviation industry (Duncan et. al, 2019). The new ECMWF IFS:

Model, created in 2021, allows industries to scale the larger impacts of turbulence in the atmosphere. Further models of IFS have advanced into machine learning models, using experimental runs in data forecasting and turbulence to predict larger scales of wind paths and velocity (Gisinger, 2024). Scientific evidence has proven that turbulence detection is being improved. Modern turbulence prediction tools–Doppler spectrum width analysis, ECMWF IFS, and STI validation–could make forecasting more precise, reducing unexpected turbulence encounters. The evolution of forecasting turbulence has been successful and has contributed to several key factors, including the quality of meteorological advancements and technological infrastructure. However, further research is required to reduce the uncertainties caused by aircraft motion. Combining the Doppler spectrum width with other radar techniques can further improve real-time turbulence forecasting in aviation.

According to Delta Airlines, the pilot protocol for turbulence was trained from day one. Pilots learn how to detect and when to anticipate turbulence during a flight. Pilots determine the potential presence of turbulence when encountered. These actions could vary from airline to airline and aircraft type (Sharman, 2016). The pilot's main priority is to keep the aircraft flying–steadying the plane during wind paths and applying pressure to maintain that the plane is lifted during a flight. Pilots tend to see turbulence under four categories: "light, moderate, severe, and extreme" (Jung-Hoon, 2011). Because of the severe turbulence and dangers during storms, pilots are required to fly around thunderstorms and avoid damage from turbulence. When turbulence occurs during a flight path, pilots can report turbulent sightings on their radar. This eventually

serves to inform the ground where other planes can be notified of recent turbulence on that jet-stream path (FAA, 2025).

In addition to technological advancements, psychological research has explored how flight anxiety is shaped by in-flight social and emotional factors. Batouei et al. examined the relationship between perceived service quality, crew competence, and flight anxiety, and found that passengers who felt reassured by confident, communicative crews reported significantly lower levels of fear. These findings suggest that passenger trust in human factors—not just technological tools—can have a measurable impact on how turbulence is experienced and interpreted (2019). Turbulence often evokes more than a physical reaction; it stirs profound psychological responses that vary across belief systems. As Parui articulates, "[passengers who are] more God-fearing, turbulence often emerges as a sign of supernatural wrath for the wrongs one had wrought… for the atheist… a more trying test of the nerves. Fear in a flight is a great leveler, capable of constructing collective paranoia in an intersubjective instant" (2016). This framing suggests that turbulence, regardless of its mechanical or meteorological causes, is often experienced as an existential disruption, a confrontation with one's mortality. Whether interpreted as divine punishment or pure chaos, turbulence taps into passenger vulnerability, which may explain why anxiety persists even when forecasting tools are highly advanced.

Methodology

Survey Distribution

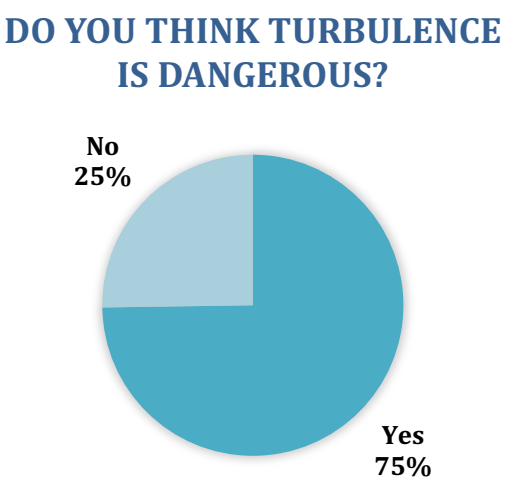
A structured online survey was distributed through SurveyMonkey and received 106 valid responses from individuals who had flown on a commercial airline within the past two years. The survey consisted of multiple-choice and Likert-scale questions designed to measure perceptions of turbulence severity, trust in forecasting tools, and behavior in response to adverse weather. Participants were recruited through public social media posts and community networks. While the survey was limited to self-selected respondents, the sample size provided a useful review of current public attitudes toward turbulence and aviation forecasting.

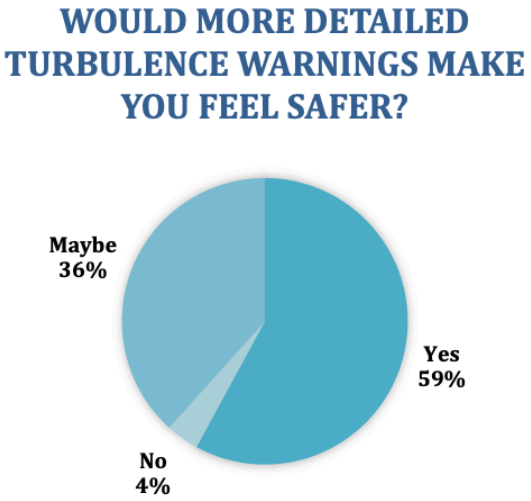
Meteorological Data Review

Passenger accounts were analyzed along with real-world meteorological data to understand how turbulence is forecasted and ultimately reported and experienced. Examined various sources including AIRMET Tango advising, Pilot Reports (PIREPs), and model outputs from the Integrated Forecasting System (IFS) of the European Center for Medium Range Weather Forecast (ECMWF) and the Spatial Turbulence Intensity (STI) analysis. Case studies were examined, such as the Delta Air Lines incidents of turbulence in 1997 and 2018 in Colorado to compare observational data and passenger perceptions. Academic literature was reviewed to analyze these results within a larger context of atmospheric science, methods of turbulence detection, and risk communication in aviation.

Results

Most participants in the survey reported experiencing turbulence during a flight, with many describing it as a stressful or fear-inducing event. As the figure shows, nearly three-quarters (74.77%) perceived turbulence as dangerous even when reassured by flight crews or safety records.





The majority of respondents (59%) said they would feel safer if given more detailed warnings about turbulence. This suggests that many passengers are looking for clear communication and reassurance during flights. Better explanations from flight crews or airlines could play a key role in reducing anxiety and building trust in the experience of flying.

Over half of the respondents said they would avoid flying near storms or mountainous regions, which supports how perception continues to influence behavior. This skepticism aligns with the findings of Keller et al. and Guarino et al., who

demonstrated that mountainous regions, such as Colorado, pose elevated and often unpredictable turbulence risks (2015; 2018).

Supporting this perception gap, meteorological data from recent turbulence events, such as the Denver International Airport flight incident in Colorado, demonstrated a clear alignment between the forecast models and observed conditions. This reinforces the reliability of modern tools, such as the ECMWF Integrated Forecasting System (IFS), Spatial Turbulence Intensity (STI), and Doppler spectrum analysis, further justifying that turbulence is often a predictable and manageable phenomenon despite public concern.

Although most survey respondents were from the South Atlantic, Pacific, and Middle Atlantic regions, only a small portion (8%) was identified in the mountainous region, which includes Colorado. Despite this, the survey's broader trends—such as viewing turbulence as dangerous and wanting more detailed warnings—may still reflect experiences on transcontinental routes**,** many of which pass over mountainous terrains. This reinforces the idea that turbulence perception is not limited to one geographic area; passengers outside the mountainous region may be affected by atmospheric conditions tied to elevation, terrain, or jet stream shifts. These insights support the need for improved communication regarding region-specific turbulence risks across all the major flight paths.

Discussion

Building on the survey results, the findings reveal a perpetual disconnect between how passengers perceive turbulence and its actual threat. Although turbulence has not contributed to a major fatal accident for over four decades, it remains one of the most misunderstood aspects of air travel. This disconnect may stem less from technical

shortcomings and more from a lack of public awareness of advancements in meteorological forecasting. Tools, such as Doppler spectrum width analysis and STI mapping, have significantly improved the accuracy and resolution of turbulence detection. However, these innovations are rarely communicated to the flying public, limiting their impact from the psychological and risk communication standpoint.

Passengers often interpret turbulence as a sign of danger or unpredictability, and this anxiety is worsened when they are not given clear information mid-flight. This emotional response is not uncommon; a 2025 survey referenced by travel journalist Eleni Weisbrot reported that more than half of surveyed travelers identified turbulence as their primary source of flight-related anxiety (Weisbrot). Weisbrot's statistic reinforces the results of this study, which suggest that passenger discomfort stems not solely from the experience of turbulence itself, but from a lack of understanding of its causes, forecasting, and aviation management.

For example, Batouei et al. (2019) found that passengers' perceptions of flight crew competence and service quality play a critical role in easing flight anxiety, further emphasizing the importance of perceived control and reassurance. Similarly, Parui offers a philosophical lens, observing that turbulence can evoke existential unease by reminding passengers of their mortality, a shared psychological experience that transcends belief systems (2016). Together, these insights highlight the need for emotional reassurance and informational clarity. Enhancing transparency around turbulence forecasting and training flight crews to communicate these concepts effectively could be the key in reducing passenger anxiety and restoring confidence in aviation safety. As this research and

previous psychological studies indicate, turbulence anxiety is often shaped more by perceived risk and informational gaps than by actual physical threats.

To better illustrate the disconnect between perception and actual causes of turbulence, the following table contrasts common passenger beliefs with scientifically validated explanations, highlighting where misunderstandings often occur:

Passenger Beliefs	Actual Causes (Forecasted/Understood by Experts)
"The wings are going to break off"	Aircraft wings are designed to withstand extreme turbulence
"We're flying into a storm"	Most turbulence occurs in clear air (CAT: Clear Air Turbulence)
"The pilots didn't see it coming"	Pilots use Doppler radar, EDR, GTG forecasts, and PIREPS
"The plane is falling"	Altitude might fluctuate, but lift and control are maintained
"Turbulence means we're in danger"	Most turbulence is mild/moderate and not a safety threat
"If there's no announcement, we're doomed"	Pilots may not announce minor turbulence to avoid unnecessary worry
"This feels worse than it is"	Perception is influenced by anxiety, not objective intensity

The table shows that passengers can hold misconceptions about turbulence that differ significantly from expert understanding and the available forecasting methods. These beliefs often stem from a lack of visible cues, limited in-flight information, or emotional interpretations of sudden aircraft movement. The persistence of these misunderstandings demands more effective communication between the aviation industry and the flying public. Providing passengers with brief, accessible explanations, either

through flight crews, pre-flight briefings, or visual aids, could help bridge this perception gap. Doing so may not only ease anxiety but also foster greater confidence in aviation safety systems, especially as turbulence incidents become more frequent in response to regional terrain effects and global climate shifts.

While this study did not include direct interviews with the aviation industry, it draws on public airline documents, model outputs, and survey responses to explore the perception gap. Airline operations have made significant advancements in forecasting and mitigating the impact of turbulence. Tom Staigle, former Captain and Chief Technical Pilot at Delta Air Lines, outlined several of these developments during a 2018 FAA Turbulence Workshop. Under Staigle's leadership, Delta transitioned from subjective pilot reports (PIREPs) to automated and objective turbulence detection using Eddy Dissipation Rate (EDR), providing consistent and reliable data. These improvements are further supported by advanced forecasting tools, such as the Graphical Turbulence Guidance (GTG) model and Domestic Nowcast, which integrate real-time radar and turbulence data. Pilots at Delta access this information through a dedicated in-flight weather app, enabling them to proactively adjust their flight paths to avoid turbulence. Despite these tools, Staigle emphasizes that "pilots are not meteorologists", and discusses the educational limitations in cockpit weather knowledge and the importance of user-friendly tools. He also raised concerns about proprietary control over weather data by vendors, advocating broader data sharing through organizations such as IATA to promote public understanding and safety. Consequently, Delta reported a measurable decline in turbulence-related injuries and improved fuel efficiency (2018).

However, passengers are often unaware of the technological sophistication of turbulence management, which contributes to the perception gap.

Bridging this gap will require not only transparency in data sharing, but also deeper exploration into how passengers interpret and emotionally respond to turbulence. Future research could expand the sample size, focus more specifically on regional turbulence (e.g., over the Colorado Rockies), and include direct comparisons between forecast data and passenger sentiment to further refine these findings. The survey revealed that a significant portion of passengers perceived turbulence as dangerous, regardless of their actual threat level. This aligns with Batouei et al.'s (2019) findings, which emphasize the importance of service quality and crew competence in shaping passengers' perceptions and reducing anxiety. Therefore, enhancing communication strategies and crew training could bridge the gap between the perceived and actual turbulence risks.

Conclusion

This study explored how passengers perceive turbulence compared with their actual forecast accuracy, revealing a clear mismatch between public anxiety and technological reliability. Although tools such as Doppler spectrum width analysis and EDR have improved turbulence prediction and flight safety, passenger understanding has not kept pace. The results suggest that the aviation industry’s communication strategies, rather than its forecasting capabilities, may be the root of this disconnect.

By examining expert insights, meteorological models, and survey responses, this study highlighted the importance of transparency in turbulence forecasting. Even with limited resources, this project demonstrates that community college-led research can

meaningfully contribute to discussions on aviation safety, meteorology, and public education. Future studies should continue to bridge the gap between public perception and atmospheric science, especially when ongoing climate trends are influencing turbulence patterns globally. Ultimately, this study emphasizes that accurate turbulence forecasting must be accompanied by clear communication strategies to ensure passengers are informed and supported throughout their flight experience.

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